

School of Computer Science and Artificial Intelligence

Lab Assignment # 15.2

Program	: B. Tech (CSE)
Specialization	: -
Course Title	: AI Assisted Coding
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Semester	II
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Task 1: Setup Fast-API Backend (VS Code) Install:

Prompt: Create a basic FastAPI app with one GET endpoint "/" that returns a welcome message.

Code:

```
from fastapi import FastAPI
app = FastAPI()
@app.get("/")
async def read_root():
    return {"message": "Welcome to the FastAPI app!"}
```

Run: python -m uvicorn main:app --reload

Output: <http://127.0.0.1:8000/>

Pretty-print ☐

```
{"message": "Welcome to the FastAPI app!"}
```

Explanation: This code sets up a basic FastAPI application with endpoints to create, read, update, and delete items in a list, along with docstrings for Swagger documentation.

Task 2: API – Create and Read Items

Prompt: Add GET `"/items"` and POST `"/items"` endpoints in FastAPI using a list to store items.

Code:

```
items = []
@app.get("/items")
async def read_items():
    return items
@app.post("/items")
async def create_item(item: dict):
    items.append(item)
    return {"message": "Item created", "item": item}
```

Output:

FastAPI 0.100 QAS 3.1
/openapi.json

default

GET

/ Read Root

GET

/items Read Items

POST

/items Create Item

Schemas

HTTPValidationError

Expand all object

ValidationError

Expand all object

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  http://127.0.0.1:8000/items/ \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{"name": "Pen", "price": 10}'
```

Request URL

http://127.0.0.1:8000/items

Server response

Code	Details
200	Response body <pre>{ "message": "Item created", "item": { "name": "Pen", "price": 10 } }</pre> Download

Explanation: This code defines two endpoints in a FastAPI application to manage a list of items, allowing users to retrieve all items and add new items to the list.

Task 3: Update Existing Item

Prompt: Add a PUT `"/items/{index}"` endpoint to update an item in a list by index.

Code:

```
@app.put("/items/{index}")
async def update_item(index: int, item: dict):
    if index < 0 or index >= len(items):
        return {"error": "Index out of range"}
    items[index] = item
    return {"message": "Item updated", "item": item}
```

Output:

The screenshot displays the FastAPI Swagger UI interface. At the top, it shows 'FastAPI 0.100.0' and 'OAS 3.1'. Below this, a list of endpoints is shown under the 'default' tag. The endpoint `PUT /items/{index}` is highlighted, with the description 'Update Item'. Below the endpoints, the 'Schemas' section shows `HTTPValidationError` and `ValidationError`. The 'Curl' section contains a successful curl command: `curl -X 'PUT' \ 'http://127.0.0.1:8000/items/0' \ -H 'accept: application/json' \ -H 'Content-Type: application/json' \ -d '{ "name": "Pencil", "price": 10 }'`. The 'Request URL' is `http://127.0.0.1:8000/items/0`. The 'Server response' section shows a '200' status code. The 'Response body' is `{ "message": "Item updated", "item": { "name": "Pencil", "price": 10 } }`. The 'Response headers' are `content-length: 62, content-type: application/json, date: Tue, 17 Mar 2026 11:01:32 GMT, server: uvicorn`. The 'Responses' section is also visible at the bottom.

Explanation: This code adds a PUT endpoint to the FastAPI application that allows users to update an existing item in the list by specifying its index, with error handling for out-of-range indices.

Task 4: Delete Item

Prompt: Add a DELETE `"/items/{index}"` endpoint to remove an item from a list.

Code:

```
@app.delete("/items/{index}")
async def delete_item(index: int):
    if index < 0 or index >= len(items):
        return {"error": "Index out of range"}
    removed_item = items.pop(index)
    return {"message": "Item deleted", "item": removed_item}
```

Output:

Curl	
<pre>curl -X 'DELETE' \ 'http://127.0.0.1:8000/items/0' \ -H 'accept: application/json'</pre>	
Request URL	
<pre>http://127.0.0.1:8000/items/0</pre>	
Server response	
Code	Details
200	Response body <pre>{ "message": "Item deleted", "item": { "name": "Pen", "price": 20 } }</pre> Response headers <pre>content-length: 59 content-type: application/json date: Tue, 17 Mar 2026 11:04:19 GMT server: uvicorn</pre>
Responses	
Code	Description

Explanation: This code adds a DELETE endpoint to the FastAPI application that allows users to remove an item from the list by specifying its index, with error handling for out-of-range indices.

Task 5: Auto-Generated Documentation

Prompt: Add docstrings to all FastAPI endpoints so they appear in Swagger documentation.

Code:

```
@app.get("/items")
async def read_items():
    """
    Retrieve all items from the list.
    """
    return items

@app.post("/items")
async def create_item(item: dict):
    """
    Create a new item and add it to the list.
    """
    items.append(item)
    return {"message": "Item created", "item": item}

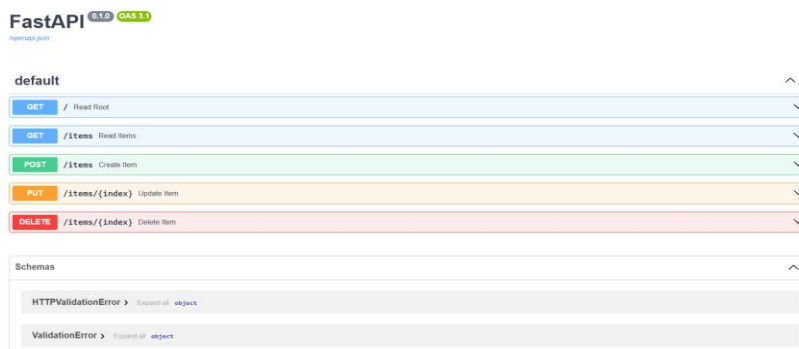
@app.put("/items/{index}")
async def update_item(index: int, item: dict):
    """
    Update an existing item in the list by its index.
    """
    if index < 0 or index >= len(items):
        return {"error": "Index out of range"}
    items[index] = item
    return {"message": "Item updated", "item": item}

@app.delete("/items/{index}")
async def delete_item(index: int):
    """
    Delete an item from the list by its index.
    """
    if index < 0 or index >= len(items):
        return {"error": "Index out of range"}
    removed_item = items.pop(index)
    return {"message": "Item deleted", "item": removed_item}
```

Open:

- <http://127.0.0.1:8000/docs> (Swagger UI)

Output:



Explanation: This code adds docstrings to each endpoint in the FastAPI application, providing descriptions that will be displayed in the Swagger UI for better documentation and understanding of each endpoint's functionality.